Preliminary Mission Analysis for a Transfer from Earth to Near-Earth Object 65803 Didymos

MAE 146: Astronautics — Project, Problem 4

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Abstract

This report presents a preliminary mission analysis for a spacecraft transfer from Earth to the near-Earth asteroid 65803 Didymos under restricted two-body dynamics. Heliocentric states of Earth and Didymos are propagated from Keplerian elements using an in-house Kepler propagator, and impulsive transfers are computed by solving Lambert's problem with universal variables under a zero-sphere-of-influence assumption. A reference transfer departing at $t_{ref}+12$ days and arriving at $t_{ref}+412$ days requires a total ΔV of 20.0300 km/s. A porkchop plot over a 1000-day window with 5-day resolution shows that substantially cheaper prograde transfers exist; the lowest-cost solution found departs at $t_{ref}+600$ days and arrives at $t_{ref}+1000$ days with a total ΔV of 7.8600 km/s, a reduction of roughly 61% relative to the reference case.

1 Introduction

Near-Earth objects (NEOs) are attractive targets for science, planetary defense, and resource-utilization missions. Didymos, the primary of the binary system targeted by the DART and Hera missions, has a moderately eccentric ($e = 0.384$) and inclined ($i = 3.41°$) orbit that crosses the region between Earth and Mars. Because its orbit differs substantially from Earth's, the cost of a rendezvous varies strongly with departure and arrival epochs, making a trade study over the launch window essential.

The objectives of this analysis are to:

1. determine the heliocentric position and velocity of Earth and Didymos at specified epochs;
2. compute and visualize an Earth–Didymos transfer for a prescribed departure and arrival date;
3. generate a porkchop plot of total ΔV over a 1000-day window; and
4. identify and plot a minimum-ΔV transfer from the porkchop plot.

2 Methodology

2.1 Problem Data

The orbital elements $\{a, e, i, \Omega, \omega, \theta\}$ at the reference epoch t_{ref} are given in Table 1, with 1 AU = 149.6×10^6 km, $\mu_\odot = 1.32712 \times 10^{11}$ km^3/s^2, and 1 day = 86400 s.

Table 1: Orbital elements at t_{ref}.

Element	Earth	Didymos
a (AU)	1.00000011	1.6442
e	0.016710212	0.38385
i (deg)	0.00005	3.4079
Ω (deg)	-11.26064	73.196
ω (deg)	102.94719	319.321
$\theta(t_{ref})$ (deg)	100.46435	195.0

2.2 Orbit Propagation

States are propagated analytically along Keplerian arcs. The initial true anomaly is converted to eccentric anomaly E via $\tan(E/2) = \sqrt{(1-e)/(1+e)}\,\tan(\theta/2)$ and then to mean anomaly via Kepler's equation $M = E - e\sin E$. The mean anomaly is advanced by $M(t) = M_0 + \sqrt{\mu/a^3}\,\Delta t$, and the inverse problem $M \to E \to \theta$ is solved with a Newton–Raphson iteration (tolerance 10^{-12}). The resulting elements are converted to inertial Cartesian state with the perifocal-to-inertial rotation $\mathbf{R}_3(-\Omega)\mathbf{R}_1(-i)\mathbf{R}_3(-\omega)$. Functions `rv2kep`, `kep2rv`, `ta2ma`, `ma2ta`, and `propagate_elliptical` implement these steps.

2.3 Lambert's Problem

Transfers are computed with the universal-variable formulation of Lambert's problem (`lambert_universal`). With $\Delta\nu$ the transfer angle (prograde, `cw`$= 0$),

$$A = \sin\Delta\nu\sqrt{\frac{r_1 r_2}{1-\cos\Delta\nu}}, \qquad y(z) = r_1 + r_2 + A\frac{zS(z)-1}{\sqrt{C(z)}}, \tag{1}$$

and the time of flight is matched by finding the root z of

$$F(z) = \frac{1}{\sqrt{\mu}}\left[\left(\frac{y}{C}\right)^{3/2} S + A\sqrt{y}\right] - \Delta t = 0, \tag{2}$$

where $C(z)$ and $S(z)$ are the Stumpff functions, evaluated with a series expansion when $|z| < \varepsilon$. A Newton iteration safeguarded by bisection is used, with tolerances $\varepsilon = \texttt{tol} = 10^{-8}$. The Lagrange coefficients

$$f = 1 - \frac{y}{r_1}, \quad g = A\sqrt{\frac{y}{\mu}}, \quad \dot g = 1 - \frac{y}{r_2} \tag{3}$$

give the terminal velocities $\mathbf{v}_1 = (\mathbf{r}_2 - f\mathbf{r}_1)/g$ and $\mathbf{v}_2 = (\dot g\mathbf{r}_2 - \mathbf{r}_1)/g$.

2.4 Delta-V Model

Assuming a zero sphere of influence at both bodies, the total impulsive cost is

$$\Delta V = \|\mathbf{v}_1 - \mathbf{v}_E(t_0)\|_2 + \|\mathbf{v}_2 - \mathbf{v}_D(t_f)\|_2. \tag{4}$$

2.5 Porkchop Grid

Departure and arrival days $t_0, t_f \in [0, 1000]$ were sampled at 5-day intervals (201×201 grid). Body states were precomputed once per grid time, and Lambert's problem was solved for every pair with $t_f - t_0 > 100$ days (prograde only). Cases where the solver failed or returned a non-finite ΔV were left blank.

3 Results

3.1 (a) Earth State at $t_{ref} + 12$ days

$$\mathbf{r}_E = [-1.376321 \times 10^8,\ -6.091012 \times 10^7,\ -7.559 \times 10^1]^T \text{ km},$$
$$\mathbf{v}_E = [11.557797,\ -27.254849,\ -2.14 \times 10^{-5}]^T \text{ km/s}.$$

3.2 (b) Didymos State at $t_{ref} + 412$ days

$$\mathbf{r}_D = [-9.335285 \times 10^7,\ 1.799321 \times 10^8,\ 8.419365 \times 10^6]^T \text{ km},$$
$$\mathbf{v}_D = [-27.465987,\ -3.475294,\ 1.505911]^T \text{ km/s}.$$

3.3 (c) Reference Transfer: $t_0 = t_{ref} + 12$ d, $t_f = t_{ref} + 412$ d

For a time of flight of 400 days the Lambert solution yields

$$\mathbf{v}_1 = [16.4780,\ -27.9105,\ -1.3396]^T \text{ km/s},$$
$$\mathbf{v}_2 = [-18.8819,\ -15.5068,\ -0.2721]^T \text{ km/s}.$$

The resulting ΔV values are summarized in Table 2, and the trajectory is shown in Fig. 1. The Lambert residual was below 10^{-6} s.

Table 2: Delta-V budget for the reference transfer.

Maneuver	ΔV (km/s)
Earth departure	5.1413
Didymos arrival	14.8864
Total	**20.0277**

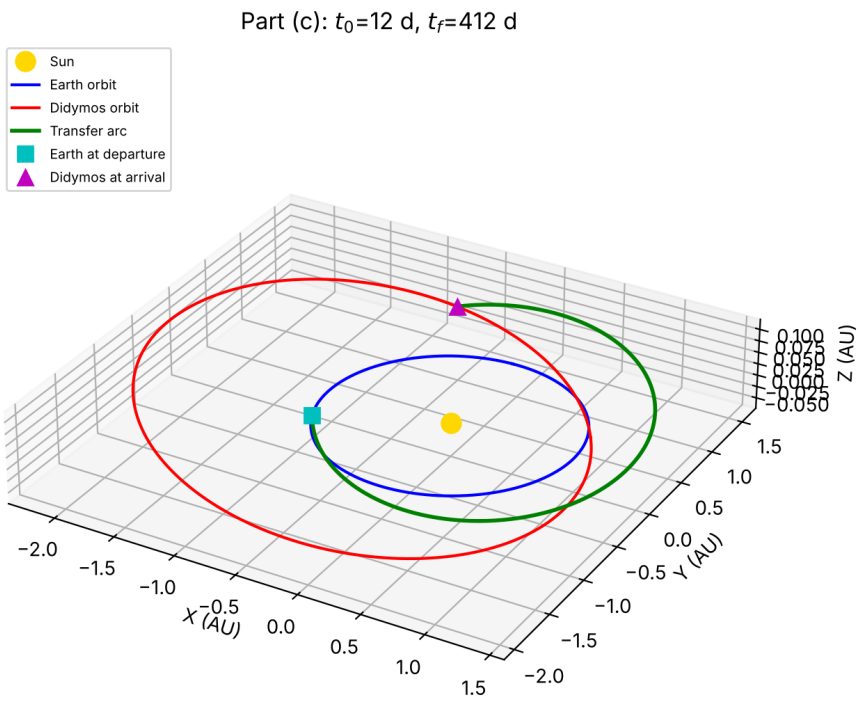


Figure 1: Reference Earth–Didymos transfer (400 days) with the orbits of Earth and Didymos.

The arrival burn dominates. Didymos is at $r \approx 2.0$ AU when the spacecraft arrives and moves in a very different direction than the transfer arc, which makes this timing poor.

3.4 (d) Porkchop Plot

Figure 2 shows the total ΔV (clipped at 25.0000 km/s) as a function of departure and arrival day. The cost is strongly periodic in departure date, reflecting the synodic relationship between Earth and Didymos, and the reference transfer in (c) lies in a high-cost region. The grid-wide minimum occurs at $t_0 = 600$ d and $t_f = 1000$ d with $\Delta V = 7.86$ km/s. The "island" structure of cheap transfers is visible near the lower-right of the feasible region.

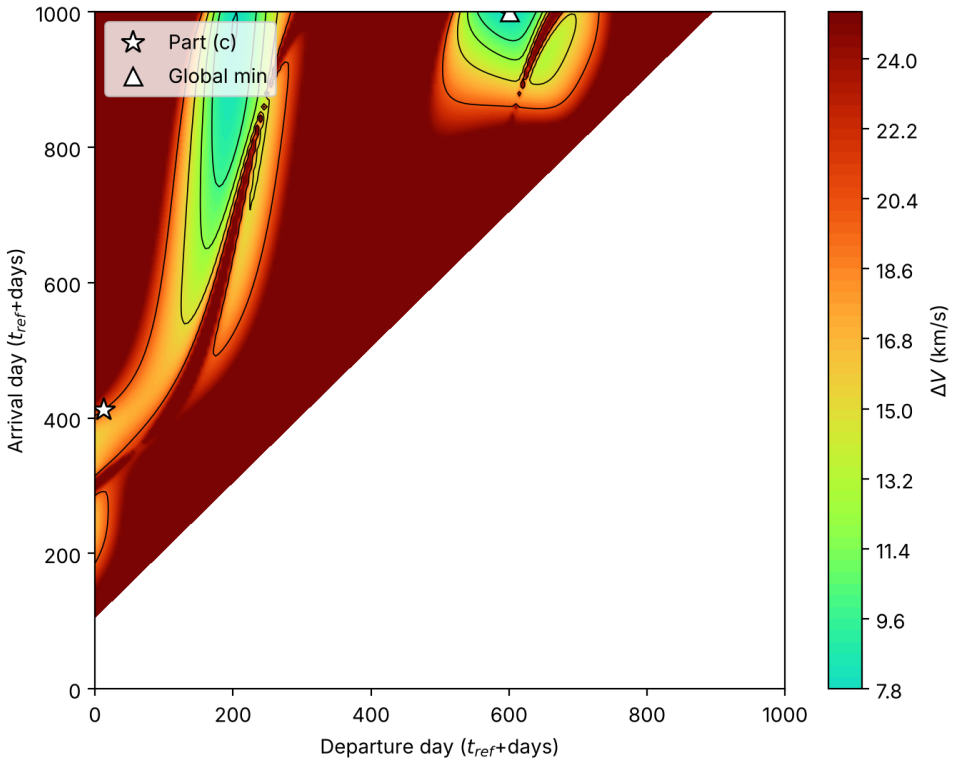


Figure 2: Porkchop plot of total ΔV for Earth-to-Didymos prograde transfers. The star marks the part (c) case and the triangle the minimum found.

3.5 (e) Minimum-ΔV Transfer

The minimum identified from the porkchop plot corresponds to

$$t_0 = t_{ref} + 600 \text{ d}, \quad t_f = t_{ref} + 1000 \text{ d}, \quad \text{TOF} = 400 \text{ d},$$
$$\mathbf{v}_1 = [-29.1318,\ 15.9409,\ 1.7637]^T \text{ km/s}, \quad \mathbf{v}_2 = [15.2961,\ 19.9151,\ -0.1894]^T \text{ km/s},$$
$$\Delta V_{total} = 7.8598 \text{ km/s}.$$

The transfer is shown in Fig. 3. Other local minima exist, for example near $(t_0, t_f) = (200, 970)$ d with $\Delta V \approx 7.96$ km/s, which is nearly as efficient and is not constrained by the edge of the search window.

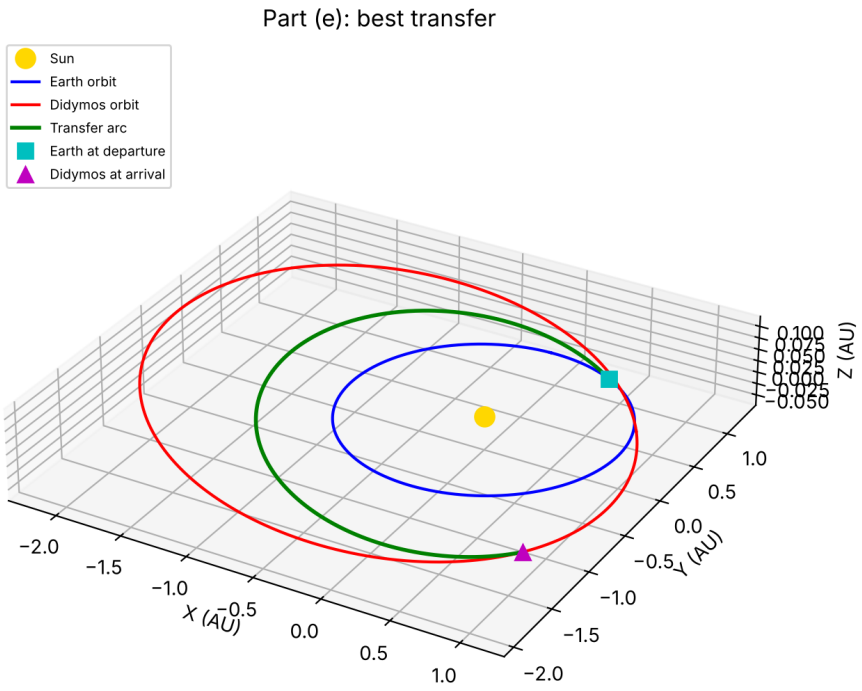


Figure 3: Minimum-ΔV transfer, with Earth's position at departure and Didymos's position at arrival.

4 Discussion

Compared with the reference case, the minimum-cost transfer cuts the total ΔV from 20.03 to 7.86 km/s, and its timing aligns the spacecraft's arrival velocity much more closely with Didymos's heliocentric velocity. Several caveats apply. First, the global minimum lies on the $t_f = 1000$ d boundary of the search window, so a longer window could reveal slightly better solutions. Second, the zero-sphere-of-influence model treats ΔV_{Earth} as the departure *hyperbolic excess* change; a real mission would add the cost of escaping Earth's gravity well from a parking orbit, and the arrival ΔV would involve a rendezvous with Didymos's very weak gravity field. Third, the model assumes unperturbed two-body motion and only prograde, zero-revolution transfers. Finally, the 5-day grid limits the precision of the optimum, and a local optimizer starting from the grid minimum would refine it.

5 Conclusion

Using Keplerian propagation and a universal-variable Lambert solver, the heliocentric states of Earth and Didymos were computed and transfers between them evaluated. The reference 400-day transfer costs 20.0300 km/s, while a porkchop search finds a transfer of 7.8600 km/s (departure $t_{ref} + 600$ d, arrival $t_{ref} + 1000$ d). Future work should include longer search windows, multi-revolution solutions, and patched-conic modeling of the planetary departure and arrival phases.

A MATLAB Script for Problem 4

Supporting functions (`rv2kep`, `kep2rv`, `ta2ma`, `ma2ta`, `propagate_elliptical`, `propagate_elliptical_ta`, `stumpff`, `lambert_universal`) are submitted separately in the code archive.

```
%% Problem 4: Mission to Didymos
clear; clc; close all;

%Constants
AU = 149.6e6; % km per AU
MU = 132712000000; % km^3/s^2
day2sec = 86400; % s/day
eps = 1e-8; % Stumpff tolerance
tol = 1e-8; % Lambert tolerance

%Define
Earth = [1.00000011*AU; 0.016710212; deg2rad(0.00005);
          deg2rad(-11.26064); deg2rad(102.94719); deg2rad(100.46435)];

Didymos = [1.6442*AU; 0.38385; deg2rad(3.4079);
          deg2rad(73.196); deg2rad(319.321); deg2rad(195.0)];

% PART (a): Earth state at t = t_ref + 12 days
fprintf("Earth at t_ref + 12 days\n");

%Find Position and Veloicty at given time by using Prop Ellpital Function
dt_a = 12 * day2sec;
rv_earth_a = propagate_elliptical(Earth, dt_a, MU); % (6x1)
r_earth_a = rv_earth_a(1:3);
v_earth_a = rv_earth_a(4:6);

fprintf("r_Earth = [%+.6e %+.6e %+.6e] km\n", r_earth_a);
fprintf("v_Earth = [%+.9f %+.9f %+.9f] km/s\n\n", v_earth_a);

% PART (b): Didymos state at t = t_ref + 412 days
fprintf("Didymos at t_ref + 412 days\n");

%Find Position and Veloicty at given time by using Prop Ellpital Function
dt_b = 412 * day2sec;
rv_did_b = propagate_elliptical(Didymos, dt_b, MU); % (6x1)
r_did_b = rv_did_b(1:3);
v_did_b = rv_did_b(4:6);

%Print Answers
fprintf("r_Didymos = [%+.6e %+.6e %+.6e] km\n", r_did_b);
fprintf("v_Didymos = [%+.9f %+.9f %+.9f] km/s\n\n", v_did_b);

%PART (c): Lambert transfer
%t0 = t_ref + 12 days (departure from Earth)
%tf = t_ref + 412 days (arrival at Didymos)
%TOF = 400 day
fprintf("Lambert Transfer (TOF = 400 days) \n");

tof_c = (412 - 12) * day2sec; % 400 days in seconds

[v1_c, v2_c, res_c] = lambert_universal(r_earth_a, r_did_b, tof_c, MU, 0, eps, tol);

dV_earth_c = norm(v1_c - v_earth_a);
dV_didymos_c = norm(v2_c - v_did_b);
dV_total_c = dV_earth_c + dV_didymos_c;

%Print Answers
fprintf("Lambert residual = %.3e\n", res_c);
fprintf("v1 = [%+.6f %+.6f %+.6f] km/s\n", v1_c);
fprintf("v2 = [%+.6f %+.6f %+.6f] km/s\n", v2_c);
fprintf("DV_Earth = %.4f km/s\n", dV_earth_c);
fprintf("DV_Didymos = %.4f km/s\n", dV_didymos_c);
```

```
fprintf("Total DV = %.4f km/s\n\n", dV_total_c);

% Transfer arc: propagate s/c from Earth Semimajor Axis, & Velocity
rv_sc0 = [r_earth_a; v1_c]; % s/c initial state (6x1)
N_arc = 500;
t_arc = linspace(0, tof_c, N_arc);
r_arc_c = zeros(3, N_arc);
for k = 1:N_arc
    rv_k = propagate_elliptical(rv_sc0, t_arc(k), MU);
    r_arc_c(:,k) = rv_k(1:3);
end

% Full orbit paths
th_v = linspace(0, 2*pi*(1-1/1200), 1200);
r_eorbit = zeros(3,1200);
r_dorbit = zeros(3,1200);
for k = 1:1200
    oe_e_tmp = Earth; oe_e_tmp(6) = th_v(k);
    oe_d_tmp = Didymos; oe_d_tmp(6) = th_v(k);
    rv_e = kep2rv(oe_e_tmp, MU);
    rv_d = kep2rv(oe_d_tmp, MU);
    r_eorbit(:,k) = rv_e(1:3);
    r_dorbit(:,k) = rv_d(1:3);
end

% 3D Plot Generation
figure('Name','Part (c): Transfer Trajectory','Color','k','Position',[50 50 950 800]);
ax = axes; hold(ax,'on'); grid(ax,'on'); axis(ax,'equal');
set(ax,'Color','k','GridColor',[0.25 0.25 0.25], ...
      'XColor','w','YColor','w','ZColor','w');

plot3(0,0,0,'yo','MarkerSize',16,'MarkerFaceColor',[1 0.9 0],'DisplayName','Sun');
plot3(r_eorbit(1,:)/AU, r_eorbit(2,:)/AU, r_eorbit(3,:)/AU, ...
      'b-','LineWidth',1.4,'DisplayName','Earth Orbit');
plot3(r_dorbit(1,:)/AU, r_dorbit(2,:)/AU, r_dorbit(3,:)/AU, ...
      'r-','LineWidth',1.4,'DisplayName','Didymos Orbit');
plot3(r_arc_c(1,:)/AU, r_arc_c(2,:)/AU, r_arc_c(3,:)/AU, ...
      'g-','LineWidth',2.2,'DisplayName','Transfer Arc');
plot3(r_earth_a(1)/AU, r_earth_a(2)/AU, r_earth_a(3)/AU, ...
      'bs','MarkerSize',11,'MarkerFaceColor','c','DisplayName','Earth at t_0');
plot3(r_did_b(1)/AU, r_did_b(2)/AU, r_did_b(3)/AU, ...
      'r^','MarkerSize',11,'MarkerFaceColor','m','DisplayName','Didymos at t_f');

xlabel('X (AU)'); ylabel('Y (AU)'); zlabel('Z (AU)');
title(sprintf('Part (c): Earth \\rightarrow Didymos \\DeltaV = %.2f km/s', dV_total_c), ...
      'Color','w');
legend('Location','best','TextColor','w','Color','k');
view(30,25);

%PART (d): Porkchop plot
%t bounbded by t_ref, t_ref+1000 days
%Skip if tf - t0 <= 100 days

fprintf(" = Part (d): Porkchop Plot =\n");

t_grid = (0:5:1000); % days from t_ref
N_g = length(t_grid);
DV_mat = NaN(N_g, N_g); % DV_mat(i,j): depart t_grid(i), arrive t_grid(j)

%Pre-compute all planet states
r_e_all = zeros(3, N_g); v_e_all = zeros(3, N_g);
r_d_all = zeros(3, N_g); v_d_all = zeros(3, N_g);
for k = 1:N_g
    rv_e = propagate_elliptical(Earth, t_grid(k)*day2sec, MU);
    rv_d = propagate_elliptical(Didymos, t_grid(k)*day2sec, MU);
    r_e_all(:,k) = rv_e(1:3); v_e_all(:,k) = rv_e(4:6);
    r_d_all(:,k) = rv_d(1:3); v_d_all(:,k) = rv_d(4:6);
end
```

```
131
132 % Lambert grid
133 fprintf("Lambert grid (%d x %d), Generating\n", N_g, N_g);
134 for i = 1:N_g
135     for j = i+1 : N_g
136         tof_days = t_grid(j) - t_grid(i);
137         if tof_days <= 100
138             continue
139         end
140         try
141             [v1_ij, v2_ij, ~] = lambert_universal( ...
142                 r_e_all(:,i), r_d_all(:,j), tof_days*day2sec, MU, 0, eps, tol);
143             dv = norm(v1_ij - v_e_all(:,i)) + norm(v2_ij - v_d_all(:,j));
144             if isfinite(dv)
145                 DV_mat(i,j) = dv;
146             end
147         catch
148             % Throws exception
149         end
150     end
151     if mod(i,20) == 0
152     end
153 end
154
155 %Global minimum
156 [~, idx] = min(DV_mat(:));
157 [ri, ci] = ind2sub(size(DV_mat), idx);
158 t0_best = t_grid(ri);
159 tf_best = t_grid(ci);
160 DV_best = DV_mat(ri,ci);
161 fprintf(" Global minimum: t0=%g day, tf=%g day, DV=%.4f km/s\n\n", ...
162         t0_best, tf_best, DV_best);
163
164 % Porkchop figure
165 figure('Name','Part␣(d):␣Porkchop␣Plot','Position',[50 50 950 780]);
166 [T0g, TFg] = meshgrid(t_grid, t_grid);
167 DV_plot = min(DV_mat', 25); % transpose + clip at 25 km/s
168
169 contourf(T0g, TFg, DV_plot, 60, 'LineStyle','none');
170 cb = colorbar;
171 cb.Label.String = '\DeltaV␣(km/s)';
172 colormap(turbo);
173 clim([0 25]);
174 hold on;
175 contour(T0g, TFg, DV_plot, [4 6 8 10 12 15 20], 'k-', 'LineWidth', 0.8);
176 plot(12, 412, 'w*', 'MarkerSize',14, 'LineWidth',2, ...
177      'DisplayName', sprintf('Part␣(c):␣%.2f␣km/s', dV_total_c));
178 plot(t0_best, tf_best, 'w^', 'MarkerSize',12, 'MarkerFaceColor','w', 'LineWidth',1.5, ...
179      'DisplayName', sprintf('Best:␣t_0=%g␣d,␣t_f=%g␣d,␣%.2f␣km/s', t0_best, tf_best, DV_best));
180
181 xlabel('Departure␣day␣(t_{ref}␣+␣days)');
182 ylabel('Arrival␣day␣␣␣(t_{ref}␣+␣days)');
183 title('Porkchop␣Plot:␣Earth␣\rightarrow␣Didymos␣␣\DeltaV␣(km/s)');
184 legend('Location','northwest');
185 set(gca);
186
187 % PART (e): Best Transfer
188 fprintf(" = Part (e): Best-Transfer 3D Trajectory =\n");
189 fprintf(" t0=%g days, tf=%g days, TOF=%g days\n", t0_best, tf_best, tf_best-t0_best);
190
191 tof_e = (tf_best - t0_best) * day2sec;
192
193 rv_earth_e = propagate_elliptical(Earth, t0_best*day2sec, MU);
194 rv_did_e = propagate_elliptical(Didymos, tf_best*day2sec, MU);
195 r_earth_e = rv_earth_e(1:3); v_earth_e = rv_earth_e(4:6);
196 r_did_e = rv_did_e(1:3); v_did_e = rv_did_e(4:6);
197
198 [v1_e, v2_e, res_e] = lambert_universal(r_earth_e, r_did_e, tof_e, MU, 0, eps, tol);
```

```
dV_e = norm(v1_e - v_earth_e) + norm(v2_e - v_did_e);
fprintf(" Lambert residual = %.3e\n", res_e);
fprintf(" Total DV = %.4f km/s\n", dV_e);

% Transfer arc
rv_sc0_e = [r_earth_e; v1_e];
N_arc_e = 600;
t_arc_e = linspace(0, tof_e, N_arc_e);
r_arc_e = zeros(3, N_arc_e);
for k = 1:N_arc_e
    rv_k = propagate_elliptical(rv_sc0_e, t_arc_e(k), MU);
    r_arc_e(:,k) = rv_k(1:3);
end

% 3D Plot Generation
figure('Name','Part␣(e):␣Best␣Transfer␣Trajectory','Color','k','Position',[100 100 950 800]);
ax2 = axes; hold(ax2,'on'); grid(ax2,'on'); axis(ax2,'equal');
set(ax2,'Color','k','GridColor',[0.25 0.25 0.25], ...
       'XColor','w','YColor','w','ZColor','w');

plot3(0,0,0,'yo','MarkerSize',16,'MarkerFaceColor',[1 0.9 0],'DisplayName','Sun');
plot3(r_eorbit(1,:)/AU, r_eorbit(2,:)/AU, r_eorbit(3,:)/AU, ...
      'b-','LineWidth',1.4,'DisplayName','Earth␣Orbit');
plot3(r_dorbit(1,:)/AU, r_dorbit(2,:)/AU, r_dorbit(3,:)/AU, ...
      'r-','LineWidth',1.4,'DisplayName','Didymos␣Orbit');
plot3(r_arc_e(1,:)/AU, r_arc_e(2,:)/AU, r_arc_e(3,:)/AU, ...
      'g-','LineWidth',2.5,'DisplayName','Transfer␣Arc');
plot3(r_earth_e(1)/AU, r_earth_e(2)/AU, r_earth_e(3)/AU, ...
      'bs','MarkerSize',13,'MarkerFaceColor','c', ...
      'DisplayName', sprintf('Earth␣(t_{ref}+%g␣d)', t0_best));
plot3(r_did_e(1)/AU, r_did_e(2)/AU, r_did_e(3)/AU, ...
      'r^','MarkerSize',13,'MarkerFaceColor','m', ...
      'DisplayName', sprintf('Didymos␣(t_{ref}+%g␣d)', tf_best));

xlabel('X␣(AU)'); ylabel('Y␣(AU)'); zlabel('Z␣(AU)');
title(sprintf('Part␣(e):␣Best␣Transfer␣␣t_0=%g␣d,␣t_f=%g␣d,␣␣\\DeltaV=%.2f␣km/s', ...
      t0_best, tf_best, dV_e), 'Color','w');
legend('Location','best','TextColor','w','Color','k');
view(30,25);
```